Geocomputation & Machine Learning course

Quick recap

The meeting focused on preparing for the final lecture and reviewing the core concepts of neural networks and convolutional neural networks (CNNs). Raffaele explained the process for selecting papers for the final discussion, requiring students to send one open-access paper each and then vote on the top two. He then recapped previous material, including the perceptron, activation functions, stochastic gradient descent, learning rates, and the use of mini-batches and backpropagation. The discussion covered different optimizers like AdaGrad and ADAM, as well as the limitations of perceptrons and the introduction of multi-layer perceptrons to handle non-linearly separable problems. Raffaele also explained regularization techniques like L1/L2 and dropout, and the concepts of underfitting and overfitting. The main part of the lecture introduced CNNs, explaining how they process image data as matrices rather than flat vectors to preserve spatial information. Raffaele detailed the convolution process using kernels, feature maps, strides, and zero padding, and explained how pooling layers reduce spatial dimensions. He described the overall architecture of a CNN, including flattening and fully connected layers for image classification, and mentioned the use of softmax for probability distributions. The session concluded with a brief discussion on model explainability using saliency maps and the possibility of using CNNs for regression tasks.

Next steps

Raffaele

- Open a poll after receiving the papers so students can vote on which ones to discuss.
- Select the top two most voted papers and prepare to explain the machine learning perspective of those papers.
- Conduct a hands-on session next Tuesday covering convolutional neural networks (CNNs), including practical material on early stopping and Hyperopt for bias and optimization.

Collaboration

- Students: Send one interesting paper each to Raffaele via email by Monday, putting Giuseppe in copy, for the last lecture discussion.
- Students: Prepare to introduce the concept and current formulation of the selected paper from a geocomputational perspective for the next Thursday lecture.

Summary

Machine Learning Lecture Planning

Raffaele announced that students should each send him one open-access paper by Monday for discussion in the final lecture, with the top two voted papers to be presented. The selected papers should focus on machine learning perspectives from a geocomputational perspective, with students explaining the problem first before Raffaele covers the machine learning aspects. The lecture then began with a review of perceptrons and activation functions, before moving into the main topic of convolutional neural networks and their application to image data.

Neural Network Loss Optimization

Raffaele discussed the concept of optimizing loss functions in neural networks, explaining how gradient descent and learning rates work to find the minimum loss. He covered different variations of gradient descent, including momentum and AdaGrad, and explained how multi-layer perceptrons can learn complex features through hidden layers. Raffaele also described back propagation and how it's used to compute gradients for training neural networks, as well as the concept of regularization techniques like dropout.

Random Forest and Model Fit

Raffaele explained the difference between Random Forest and their current approach, emphasizing how Random Forest focuses on a single important feature during training but can struggle with test-time precision if that feature is unreliable. He then discussed the concepts of underfitting and overfitting, using visual examples to illustrate how underfitting occurs when models are too simple for the task while overfitting happens when models learn the noise in training data and cannot generalize well to new data. Raffaele announced upcoming practical sessions on early stopping and Hyperopt to help prevent overfitting and optimize model parameters.

Neural Network Debugging Techniques

Raffaele discussed debugging techniques for neural networks, explaining that common issues include underfitting, overfitting, and using inappropriate features. He emphasized the importance of making hypotheses about problems and testing solutions like changing learning rates, using different seeds, or modifying model architecture. The discussion then shifted to convolutional neural networks (CNNs), where Raffaele explained the difference between processing images as matrices versus flattening them into vectors, highlighting that spatial information is lost when images are vectorized. He noted that while flattening images works for simple patterns, CNNs are necessary for complex images due to their ability to extract local patterns from images.

Convolution Neural Networks in Spatial Analysis

The team discussed the potential benefits of using convolution neural networks in a spatial context, with Andrea suggesting they could address spatial dependence. Giuseppe explained how convolutional neural networks introduce information about nearby pixels, which is important for understanding geographical relationships. Raffaele indicated they would cover more details about convolutions and convoluted features after the break, and the meeting was scheduled to resume at 15:00.

Convolutional Neural Networks Image Processing

Raffaele explained how convolutional neural networks handle image inputs, describing the process of loading images from directories with corresponding annotation files. He detailed the convolution process using a 3x3 kernel example, explaining how filters detect features through point-wise multiplication and how this creates feature maps. Raffaele noted that handling large image datasets requires additional data fetching steps due to computational intensity, and he addressed Ali's question about TIFF layers by indicating he would answer this question later in the session.

Convolution Process and Image Features

Raffaele explained the convolution process to Andrea, demonstrating how to perform pointwise multiplication and summing operations on a 3x3 kernel to analyze image features. He clarified that in photorealistic images, input would be a tensor with shape 3 (RGB channels) times height times width, and explained that convolution parameters like kernel size and stride

can be adjusted to control feature extraction precision. Raffaele also discussed how raster data with multiple environmental variables could be processed by patchifying large images into smaller manageable pieces for training deep learning models.

Convolutional Neural Network Architecture

Raffaele explained how convolutional neural networks extract color layers by using tensors with specific channel configurations for red, green, and blue components. He detailed the process of convolution operations, including zero padding, kernel sizes, and how spatial dimensions are reduced through pooling layers to compress features while preventing overfitting. The discussion covered the transition from feature maps to vector representations through flattening, followed by fully connected linear layers that produce probability distributions for classification tasks. Raffaele clarified that while the programmer designs the network architecture including convolutional layers, filters, pooling operations, and connections, the actual weights within kernels are optimized during training through backpropagation.